DECISION BRIEF · STUDIO-STANDARD-AGENT-FRAMEWORK · 25 AUG 2026

Framework Without a Runtime

The rules document is the best thing either studio repo contains. The container around it is two commits old, unlicensed, unversioned, and already forked against the framework it is meant to interoperate with.

Scope full read, both repos **Head** 078593e **Writes** none **Decisions** 15 · ~21 engineer-days

Full report Utopia-Agent-Framework-Review.pdf

4

OUT OF 10

The content would score 8. The container scores 2. `atelier-learnings` is 38 rules, each traceable to a dated production failure — and nothing comparable ships in any of the eight frameworks I compared it against. That is the real asset.

But the repo it sits in is 48 files, 44 of them Markdown, with **zero lines of executable code.** No package manifest, no CI, no `LICENSE`, no test a machine can run. It is a prompt-and-governance pack for Claude.ai Skills — not a framework in the sense LangGraph, CrewAI, Mastra or Flue are. **There is no runtime to review, and no agent loop anywhere in the studio.**

You asked for a standardised, modular, set-in-stone framework. This is a very good draft of the *specification* half of one, and none of the runtime half.

The five findings that matter

1 ### STATE-1 makes the harness bake-off unwinnable

"Resume from the Convex log alone; if resume needs the vendor file, the design fails" disqualifies **every** harness that owns its own state store. Mastra failed the gate-6 probe for exactly that. Flue's "durable stream" is its own store, so it will fail identically. LangGraph writes to its own checkpointer; CrewAI has its own backends. **As written, the bake-off has a predetermined answer of "none."** Add the adapter clause — and note it now has a concrete instance: `@mastra/convex` **puts Mastra's threads, messages and workflow snapshots in your own Convex schema**. **Your gate-6 probe used Mastra's default LibSQL store; it measured a configuration, not the framework.** Half a day re-running the kill-test with `ConvexStore` closes the bake-off either way.

2 ### The Hermes headline is materially misleading

The README advertises *"6 pass · 1 partial · 0 fail, 24/25 on the rubric."* The 5 adversarial cases are not in the results table at all; `A3` and `A4` are annotated *"observed: FAILED, uncorrected in-skill."* `gate_integrity` scored **5/5** while both uncorrected regressions are gate-integrity failures. For a pack whose first principle is that nothing grades its own work, the scoring boundary was drawn around the passing cases. `RESULTS.md` admits this in its Open Items; the README does not, and the README is what people read. **This is a 15-minute fix and the highest-credibility item on the list.**

3 eval-first-spec has already forked, with different pass bars

`studio-product-framework/agents/context/discovery/eval-first-spec.md` says **"target 20; minimum 5 to start commit."** The skill says **20, fewer is auto-fail**, with a ≥14 `[Fact]` floor and a fixed 6/7/4/3 band spread the fork has none of. The L2 autonomy rung also differs — "acts on approval" versus "act in narrow band + audit" are different blast radii. A fellow filling the product template clears the commit gate at five cases; the same artefact auto-fails the standard. **Both are called the studio standard.**

---

4 Three contradictions between the stages and the rules layer

Memory: `agent-design` Step 3 mandates `lessons.md` — a prose file — while `MEM-3` forbids prose blobs, `MEM-7` demands tenant-keyed indexed tables and `STACK-1` names Convex as sole source of truth. **Surfaces:** `workflow-design` Step 5 assigns runtime to Claude.ai / Claude Code / Cowork, which `agent-prd` Appendix C explicitly classifies as authoring-only. **Eval bar:** stage 3 requires 20 cases, stage 4 requires 10 — so **the last gate before work orders is looser than the kill line three stages earlier.** Plus nine dangling skill references, two of which are blocker exits that route a fellow to a skill they do not have.

---

5 There is no agent loop anywhere in the studio

`startAgentRun` is exported from `@studio/ai-runtime` and called from nowhere. `agentRunStarted` / `Succeeded` / `Failed` are defined and emitted nowhere. `gateway.ts` says it outright: *"One round only... No multi-hop agent loop."* And `apps/web/convex/schema.ts` has no `agentRuns` and no `agentEvents` table — which means **`STATE-1`'s kill-test currently has no target to run against.** Neither Mastra's failure nor Flue's eventual result is being measured against real studio infrastructure.

> WHAT TO PROTECT, NOT REBUILD
>
> The `atelier-learnings` rule corpus · the rung ladder and its default-down rule · `agent-prd`'s Gate 1B run-sequence table · the eval contract format (20 cases, 6/7/4/3 spread, 14-`[Fact]` floor, derived failure rates, cost-per-outcome) · the Hermes judge protocol, especially the bare-model run on every model release as a deprecation signal. **None of the eight frameworks compared ships any of these.** Positioned as the governance and eval layer *over* an adopted runtime, this is genuinely ahead. Positioned as a competing framework, it loses on eight of nine dimensions.

Scorecard

Architecture	5/10	Coherent 3-layer intent; three unreconciled layer violations; one file is 34% of the corpus
Agent abstraction	3/10	Three competing taxonomies; misses the builder/runtime split the product framework calls binding. "Sandbox" appears zero times
Modularity	3/10	*"Install all six. They are a chain, not a menu"* — anti-modular by its own statement; no package, no version, no import
Extensibility	3/10	Adding a stage touches 7–10 files, five shared. Adding a *rule* is excellent — that part alone is a 9
Observability	2/10	Mandates Langfuse from commit one; ships zero tracing of itself. Evidence base is screenshots
Testing	4/10	Real rubrics and adversarial cases — all manual, 2 of 6 skills unharnessed, three rubrics byte-identical, known failures excluded from the headline
Documentation	7/10	Best dimension. Clear, honest about the undecided harness, and the S1–S7 smoke table is a real ten-minute acceptance test
Production readiness	2/10	Nothing runs. `STATE-1`'s kill-test has no target schema to run against
SOTA competitiveness	3/10	Not a runtime, so not competitive on the runtime axis. Ahead on the governance axis, which is not in the table

Build · adopt · wrap

COMPONENT	VERDICT	WHY, LICENCE, LOCK-IN
`atelier-learnings`, rung ladder, Gate 1B, eval contract, Hermes protocol	BUILD — ALREADY BUILT	Nothing comparable in any of the eight comparators. This is the actual asset — protect it and version it.
Agent loop + run state	ADOPT MASTRA + @MASTRA/CONVEX	Apache-2.0 outside `ee/`. `ConvexStore` puts threads, messages and workflow snapshots in your own `convex/schema.ts`, indexed. `requireApproval: true` pauses *between tool selection and execution* — `LOOP-5`, which most harnesses cannot do. Use its **Workflow** primitives, not the opaque agent loop, so `LOOP-1` holds. Caveat: authenticates as Convex admin, so tenant isolation is application code's job. Fallback: `@convex-dev/agent`.
Tracing, cost accounting	ADOPT — ALREADY CHOSEN	Langfuse is already implemented in `packages/observability`. Wire it; don't re-decide it.
Durable execution, retries	ADOPT — ALREADY CHOSEN	`STACK-2` names Inngest / Trigger.dev — but neither string appears anywhere in the product framework. Convex Workflow is the lower-friction third option given `STACK-1`.
Graph orchestration	WRAP, LATER	LangGraph, MIT. Only once someone writes down what fails with a single loop (`TOOL-2`). Python service behind an HTTP boundary. Avoid LangSmith — you have Langfuse.
Own append-only `agentEvents`	BUILD — ALONGSIDE MASTRA	Mastra stores a workflow *snapshot*. That resumes, but it is not `LOOP-2`'s per-iteration event and it is not a trace archive. Two tables: Mastra's snapshot is canonical for *resume*, yours is canonical for *audit and evals*.
Flue	PROBE, EXPECT THE SAME RESULT	Apache-2.0, 8.0k★ — but created Feb 2026, no tagged releases, last push 8 Aug. Its "durable stream" is its own store.

COMPONENT	VERDICT	WHY, LICENCE, LOCK-IN
CrewAI	SKIP	Role/goal/backstory crews are the multi-agent-by-default pattern your own `TOOL-2` argues against. Adopting it would contradict your rules layer.

Decisions — ordered by impact to effort

#	RECOMMENDATION	IMPACT	DAYS	RISK IF SKIPPED	Y/N
1	Correct the Hermes headline in `README.md`; add the 5 adversarial rows (incl. A3/A4 failing) to `hermes/RESULTS.md`	HIGH	0.25	A pack built on "nothing grades its own work" publishes a self-scored result that hides its two known failures	☐
2	Add a `LICENSE` to `studio-standard-agent-framework`	HIGH	0.25	Public repo, no licence = all rights reserved. Nobody may legally reuse it, including other Utopia entities and clients	☐
3	Fix the eval-bar contradiction — `agent-prd`'s hard gate cites stage 3's contract instead of restating "ten"	HIGH	0.5	The last gate before work orders is looser than the kill line three stages earlier. Gate inversion	☐
4	Add the `STATE-1` adapter clause — or declare `STATE-1` absolute — before any further bake-off probe	VERY HIGH	1	The bake-off cannot terminate. Flue's probe will "fail" for a reason that has nothing to do with Flue	☐
5	**Re-run the `STATE-1` kill-test against Mastra +** `@mastra/convex`; keep `@convex-dev/agent` as fallback	VERY HIGH	0.5	The bake-off is blocked on a result that measured the default LibSQL store, not the framework	☐
6	Resolve the fork — one `eval-first-spec`, one pass bar, one autonomy ladder	VERY HIGH	1	Two "studio standards" with different bars and different L2 semantics, in two repos, already live	☐
7	Add `kind: builder | runtime` as a required Gate 0 field, with sandbox + `maxTurns` + `maxSpendCredits` mandatory when runtime	VERY HIGH	1	The chain emits PRDs for user-facing agents with no sandbox and no spend ceiling — the two things `@studio/ai-runtime` hard-fails without	☐
8	Rung-condition the memory model in `agent-design` Step 3; fix Appendix C's three-name/four-tier error	HIGH	1	Following stage 2 exactly violates `MEM-3`, `MEM-7`, `MEM-8` and `STACK-1` at once, and passes every gate	☐
9	Resolve the nine dangling skill references — vendor them, or mark external with a URL and a fallback line	HIGH	1	Two of them are blocker exits. The chain stops and routes a fellow to a skill they do not have	☐
10	Retitle `workflow-design` Step 5 to "Authoring surface"; add a required adjacent `HOME-1` runtime-home field	MEDIUM	0.5	Stage 1 hands stage 4 a surface built from a vocabulary the rules layer classifies as authoring-only	☐
11	Version the pack — `VERSION`, `CHANGELOG.md`, `pack_version` frontmatter, a git tag	HIGH	1	"Set in stone" with no version. Two fellows run different rules with no way to detect it	☐
12	Define an `AgentManifest` JSON Schema as the chain's machine-readable output, alongside the `.docx`	VERY HIGH	3	The chain's only output is an unparseable document. Nothing downstream can gate on it, diff it, or count it. The keystone interop artefact	☐
13	Add `agentRuns` + `agentEvents` to `schema.ts`; emit `agentRunStarted` from `startAgentRun`	VERY HIGH	4	`STATE-1`'s kill-test has no target. `STACK-4` is an assertion, not a fact. Dead code stays dead	☐
14	Build `harness/run.ts` — automated case runner plus separate-judge scorer emitting scored JSON	HIGH	5	26 manual cases across 4 harnesses means the "full set on every edit, twice on every model release" cadence will not happen	☐
15	Extract `_shared/` — memory table, evidence ladder, kill line, rung ladder, gen≠eval — and link instead of restating	MEDIUM	2	Five concepts copied across 3–4 files each. Every copy is a drift point	☐

IF YOU ONLY DO ONE WEEK

Items 1–7 are 4.5 engineer-days and they close every credibility and correctness defect: the misleading headline, the missing licence, the gate inversion, the unwinnable bake-off, the missing front-runner, the live fork, and the missing sandbox/spend gate. Items 12–14 are the real build (12 days) and can wait for a decision on team capacity.

THE ASSUMPTION MOST LIKELY TO BE WRONG

You left the team-size placeholder unfilled. I inferred **two engineers plus non-technical fellows** from the contributor graph — 2 on the product repo, 1 on the standard — and from the rung ladder's explicit non-builder path. That drives items 12–14. **A two-person team cannot maintain a six-document pack, a harness bake-off, and a test runner without cutting one of them.** If that number is wrong, tell me and the bottom third of this table changes.

Mastra rows corrected 25 Aug after reading mastra-ai/mastra source; see HARNESS-DECISION.md. · Nothing was written, committed, or pushed to either repository. Both were cloned read-only and left untouched. · Full 21-page report, with the two architecture diagrams and per-file evidence: Utopia-Agent-Framework-Review.pdf